REVIEW · REVIEW

Week 2 recall day

Practice questions · Day 16 · Sun 8 Nov · pack R2

Part A — Machines short-answer bank

A.1 Past-paper short answers (all with official model points)

A1. [Jun23 Q4c(i), 3 marks] Explain the concept of power-factor correction. Why is power-factor correction useful in power systems?

A2. [Jun23 Q4c(ii), 3 marks] Carefully explain the differences and similarities between transformers and induction machines.

A3. [Jun23 Q4c(iii), 2 marks] Consider a squirrel-cage induction machine. If the rotor cage was made from a perfect conductor (with infinite conductivity), how would this affect the properties of the machine?

A4. [Jun24 Q4c(i), 3 marks] We have discussed three types of electric power: real, reactive and apparent. Explain briefly what each one is and their practical significance.

A5. [Jun24 Q4c(ii), 3 marks] Consider a transformer operating from an AC supply. Carefully explain the main types of losses in the transformer, how they are modelled in the simplified equivalent circuit, and how they vary with output power.

A6. [Jun24 Q4c(iii), 3 marks] Carefully explain why the input current of an induction machine under no-load conditions is not zero. What are the physical effects which cause this?

A7. [2023 Quiz Q7, 4 marks] In designing a transformer using a given magnetic core, what is the process for selecting the number of turns on the primary winding for a given primary voltage? Carefully explain the results of using too few turns. How about if too many turns are used?

A8. [2024 Test Q8, 4 marks] Consider an ideal transformer driving a resistive load of resistance R. Carefully explain what would be the effect on the output voltage and current if the load resistance was doubled. Carefully explain why and how this would be different for a real transformer.

A9. [Jun23 Q2c(iii), 2 marks] For a transformer with a purely resistive load, would you expect the referred secondary voltage to be leading, in phase or lagging with respect to the primary voltage? Carefully explain why.

A10. [Jun23 Q3a(i)–(iii), 4 marks] A 50 Hz AC machine has a rated speed of 560 rpm. How many poles does it have? Is it synchronous or induction? Why? What is its ideal efficiency at rated speed?

A11. [Jun24 Q3a(ii), 3 marks] For a synchronous machine, a DC machine and an induction machine, indicate how you could halve the no-load (synchronous) speed while ensuring safe machine operation.

A12. [Jun24 Q2c(iii), 2 marks] A transformer is in the open-circuit condition. If the input voltage is halved, by what factor would the input current magnitude, input real power, input reactive power and input power factor change? (Give the reasoning, not just the factors.)

A.2 Likely new short answers on Ch4–8

N1. [Lecture 7.1.2, 7.1.4; 3 marks] Explain how an induction motor produces torque, and why it cannot produce torque at synchronous speed.

N2. [PQ short-answer (b), Lecture 7.7; 3 marks] Explain the tests used to determine the equivalent-circuit parameters of an induction machine.

N3. [PQ short-answer (c), Lecture 7.6.1; 3 marks] Explain briefly (no formulas) how the equivalent circuit of an induction machine can be used to calculate its electrical and mechanical performance at a given speed.

N4. [PQ short-answer (d), PQ 7.10(k), Lecture 7.9.1–7.9.2; 3 marks] Name the two types of induction motor rotor construction and the main difference between them. Why might external rotor resistance be added?

N5. [PQ short-answer (f), Lecture 7.9.4–7.9.5; 3 marks] Explain the principles of volts-per-hertz control of induction machines.

N6. [PQ short-answer (g), Lecture 8.1.2, 8.2.2–8.2.4; 3 marks] Using the concept of rotating magnetic fields, explain briefly how a synchronous machine operates.

N7. [PQ short-answer (h), Lecture 8.1.4–8.1.5; 2 marks] Why are salient-pole synchronous machines normally used for low-speed generators, while round-rotor machines are used for high-speed generators?

N8. [PQ short-answer (j), Lecture 8.6.5; 3 marks] Explain what limits the reactive power capability of a synchronous machine.

N9. [PQ short-answer (k), Lecture 8.6.6; 3 marks] Why is the power factor of industrial plants usually lagging? Why are synchronous motors sometimes used in such plants?

N10. [PQ 7.10(q) and 7.11(d), Lecture 7.4.5, 7.5.2; 3 marks] The effective rotor resistance can be thought of as two separate resistances. Explain their physical meaning. Hence explain the ideal efficiency of an induction motor and why the actual efficiency is lower.

N11. [Lecture 8.3.1, 8.3.4; 3 marks] What are the advantages of the per-unit system for three-phase machine and power-system calculations?

N12. [Lecture 8.5.2–8.5.4; 3 marks] Describe the open-circuit and short-circuit tests on a synchronous machine. Why does the short-circuit curve show no saturation?

Part A scoring

Count your ✓, ~ and × in A.1 (12 parts) and A.2 (12 parts). For every ~ or ×, write the one point you missed on your reference sheet under "Q4c points". Many of these points reappear across questions: reactive power as oscillating energy, copper loss proportional to load squared, and slip-frequency rotor currents.

Part B — Mixed recall

B1. [M · M7] A 6-pole, 50 Hz induction motor runs at 960 rpm. Find the synchronous speed, slip, rotor (slip) frequency and ideal efficiency.

B2. [D · D5] Find the 4-point DFT of $x(n) = \{\underline{1}, 2, 0, 1\}$.

B3. [E · E6] A BJT with $\beta = 100$ is biased at $I_C = 1$ mA. Find g_m, r_e and r_π. (WS6 Q1.)

B4. [M · M6] A PM DC motor has $k = 0.15$ V/(rad/s) and $R = 0.4\ \Omega$. On a 24 V supply it draws 5 A. Find the torque, speed in rpm and efficiency.

B5. [D · D5] Find the 4-point circular convolution of $x_1(n) = \{\underline{1}, 0, 2, 1\}$ and $x_2(n) = \{\underline{1}, 1, 0, 0\}$.

B6. [E · E6] **Method, then numbers.** Voltage-divider CE amplifier: $V_{CC} = 12$ V, $R_1 = 39$ kΩ, $R_2 = 8.2$ kΩ, $R_C = 3.3$ kΩ, $R_E = 1$ kΩ fully bypassed, $\beta = 100$, $r_o = \infty$. What is the first step? Then find r_e, Z_i, Z_o and A_v.

B7. [M · M9] **Method.** You are given a 415 V, 50 Hz, star-connected induction machine's simplified equivalent-circuit parameters and an operating speed, and asked for the torque and input current. List the first four steps.

B8. [D · D5] **Method.** You want the linear convolution of a length-5 sequence with a length-3 sequence using a radix-2 FFT. What DFT size do you use? What goes wrong with $N = 6$?

B9. [E · E6] Same amplifier as B6, but with R_E **unbypassed**. Find Z_b, Z_i and A_v. Why is this gain more predictable?

B10. [M · M8] An induction machine has $R_1 = 0.5\ \Omega$, $X_1 + X_2' = 3\ \Omega$ and $R_2' = 0.6\ \Omega$. Find the slip at maximum torque. What external referred rotor resistance gives maximum torque at standstill?

B11. [D · D6] An analogue signal has bandwidth $B = 4$ kHz. You want a radix-2 DFT with resolution ≤ 25 Hz. Find the minimum sampling rate, the number of samples and the record length. (Same method as T5 Q5.)

B12. [E · E7] Emitter follower: $R_B = 220$ kΩ, $R_E = 2.2$ kΩ, $\beta = 100$, $r_e = 10\ \Omega$. Find Z_i, Z_o and A_v.

B13. [M · M8] Locked-rotor test on a star-connected induction motor: 100 V (line), 10 A, 600 W (total). The DC test gives $R_1 = 1.2\ \Omega$ per phase. Find R_2' and X.

B14. [D · D6] A 256-point DFT of a real signal sampled at $F_s = 8$ kHz has magnitude peaks at $k = 40$ and $k = 216$. What frequency is present? If you zero-pad to 1024 points, does the resolution improve?

B15. [E · E7] **Method.** Rank the CE (bypassed), emitter-follower (CC) and common-base configurations on Z_i, Z_o, $|A_v|$ and phase.

B16. [M · M10] A generator is rated 100 MVA, 11 kV. Find the base current and base impedance. Express a 1.5 Ω per-phase reactance in per-unit.

B17. [D · D7] Find $X(z)$ and its ROC for $x(n) = (0.5)^n u(n) - 2^n u(-n-1)$. Does its DTFT exist?

B18. [E · E7] An amplifier has $A_{vNL} = -200$, $Z_i = 2$ kΩ and $Z_o = 4$ kΩ. It is driven from a source with $R_s = 1$ kΩ and loaded with $R_L = 4$ kΩ. Find A_{vL} and A_{vs}.

B19. [M · M11] A round-rotor generator has $X_S = 1.2$ pu (neglect R). It delivers rated VA at 0.8 pf lagging, at $V = 1$ pu. Find E_0, the rotor angle, and the maximum power at this excitation.

B20. [D · D7] **Method.** $H(z)$ has poles at $z = 0.5$ and $z = 1.2$. List the possible ROCs and say for each whether the system is causal and whether it is stable.

B21. [E · E8] Two RC-coupled stages. Stage 1: $A_{v1,NL} = -150$, $Z_{o1} = 3$ kΩ. Stage 2: $Z_{i2} = 1.5$ kΩ, $A_{v2,NL} = -120$, $Z_{o2} = 2.2$ kΩ. The load is $R_L = 10$ kΩ. Find the overall gain. What is the common mistake?

B22. [M · M8] A 4-pole, 50 Hz induction machine with $R_2' = 0.4\ \Omega$ is driven at 1530 rpm. Find the slip, R_2'/s and the rotor frequency. Motor or generator? What sign do the airgap power and torque have?

B23. [D · D8] **Method, then numbers.** Find $x(n)$ for

$$X(z) = \frac{1}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})}, \qquad |z| > 0.5$$

B24. [E · E9] (a) An n-channel JFET has $I_{DSS} = 10$ mA and $V_P = -4$ V. Find I_D at $V_{GS} = -1.5$ V, and V_{GS} for $I_D = 2.5$ mA. (b) An n-channel E-MOSFET has $k = 0.3$ mA/V² and $V_T = 2$ V. Find I_D at $V_{GS} = 5$ V in saturation, and the minimum V_{DS} for saturation.

Part B scoring

Count your ✓, ~ and × for each subject:

Subject	Questions	✓	~	×
Machines	B1, B4, B7, B10, B13, B16, B19, B22			
DSP	B2, B5, B8, B11, B14, B17, B20, B23			
EDC	B3, B6, B9, B12, B15, B18, B21, B24			